

PHYS 230 - Electricity and Magnetism

Lecture 5

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

- 1 Electric Charge and Electric Field
 - Charge Distributions (Continued)

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

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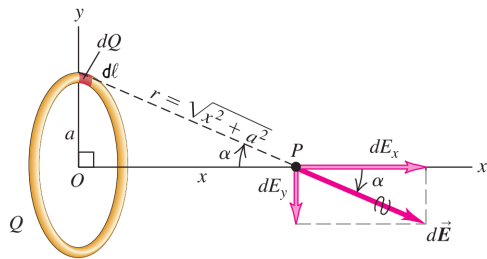
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Exercise I

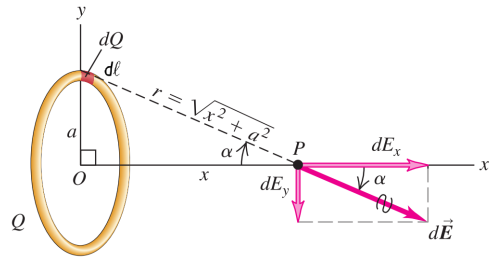
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A ring of radius a carries a uniformly distributed positive total charge Q . Calculate the electric field due to the ring at a point p lying a distance x from its center along the central axis perpendicular to the plane of the ring.



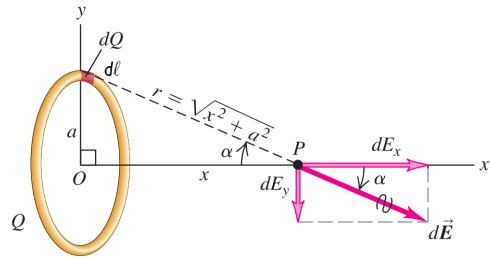
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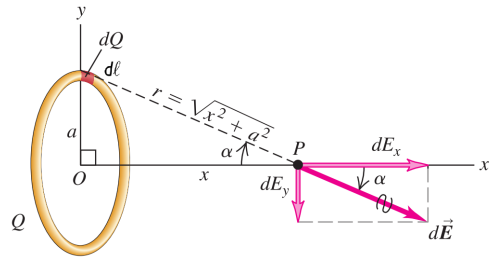
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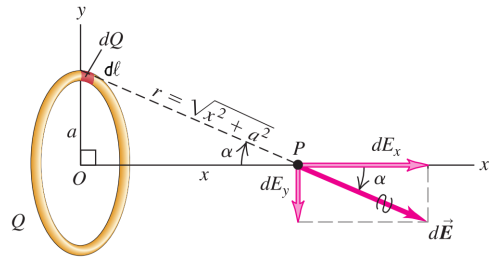
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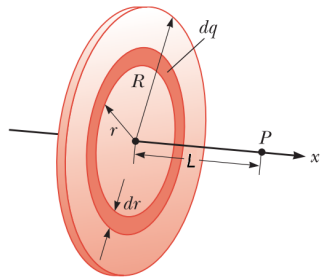
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Exercise 2

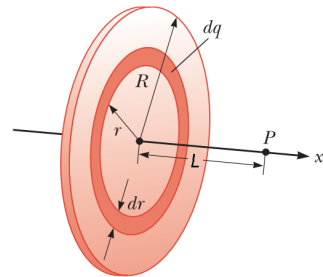
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A disk of radius R has a uniform positive surface charge density σ . Calculate the electric field at a point p that lies along the central perpendicular axis of the disk and a distance L from the center of the disk. What if we have an infinite sheet?



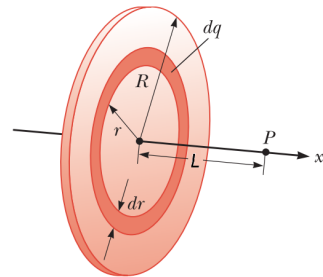
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